UNEQUAL CONNECTIONS

Deprivation and proximity to rail-based public transport across London

GEOSPATIAL ANALYSIS | 4,994 LSOAs | IMD 2025 + TfL STATION PROXIMITY

RESEARCH QUESTION

Where do areas of high socioeconomic deprivation coincide with poor proximity to rail and Underground stations across London?

RATIONALE

London has an extensive rail-based public transport network, yet access to stations and socioeconomic disadvantage are not distributed uniformly across the city. Identifying where these patterns overlap is important because neighbourhoods experiencing both higher deprivation and weaker rail proximity may face additional barriers to employment, education, services and other opportunities. This study therefore examines deprivation and rail-station proximity together at LSOA level, allowing neighbourhood-scale variation to be identified rather than hidden within larger administrative areas. The aim is to locate areas where higher deprivation coincides with poorer proximity to rail-based public transport and to provide a clear spatial evidence base for further accessibility assessment and targeted planning.

SOURCE DATA

The analysis used three authoritative datasets: English Indices of Deprivation 2025 (IoD2025) [1], ONS 2021 Lower Layer Super Output Area (LSOA) boundaries [5], and current TfL rail-based station data obtained from Open Data / Unified API sources [6]. IMD provides a consistent neighbourhood-level measure of relative deprivation across England, with decile 1 representing the most deprived 10% of LSOAs nationally and decile 10 the least deprived 10%.

The ONS LSOA geography was used as the common spatial unit because it supports detailed neighbourhood-scale comparison across London. After filtering the national boundary dataset to Greater London, 4,994 LSOAs were retained. IMD records were joined using the unique LSOA21CD identifier, with all London records successfully matched and no missing IMD attributes.

TfL station locations were used to represent access to the rail-based public transport network, including Underground, DLR, London Overground, Elizabeth line and National Rail services. All spatial datasets were projected to British National Grid (EPSG:27700) to enable consistent planar distance measurement in metres.

METHODOLOGY

IMD 2025 attributes were joined to 2021 LSOA polygons using the LSOA21CD identifier. LSOAs were selected because the English Indices of Deprivation are reported at this neighbourhood-scale geography, allowing local variations in deprivation to be examined consistently across London [1]. One representative point constrained within each LSOA was then used to calculate planar distance to the nearest rail-based station in British National Grid (EPSG:27700).

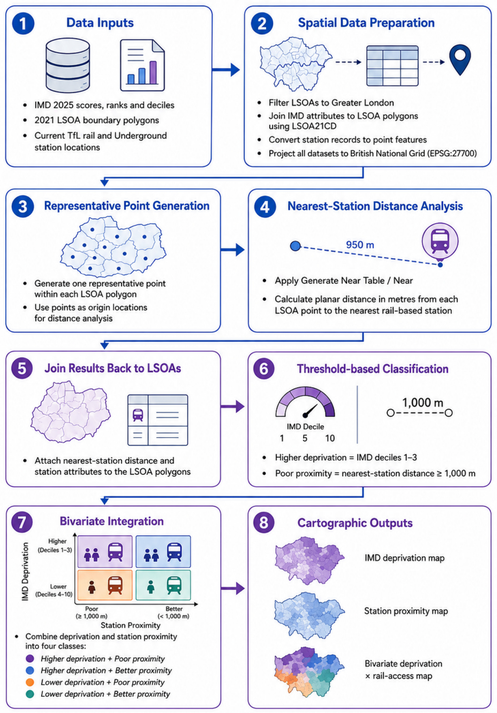
Nearest-station distance was selected as a simple and interpretable indicator of spatial proximity to the rail network, although accessibility is multidimensional and cannot be fully represented by distance alone [2]. LSOAs within IMD national deciles 1–3 were classified as higher deprivation.

Poor station proximity was defined as a nearest-station distance of ≥1,000 m. This threshold was selected because the observed 75th percentile of the London-wide distance distribution was 953 m, making 1 km a transparent rounded threshold. It also closely corresponds to the 960 m maximum walking distance used by TfL for rail and Underground services within its PTAL methodology [3]. Sensitivity tests at 750 m and 1,250 m were used to examine how the classification changed under alternative thresholds.

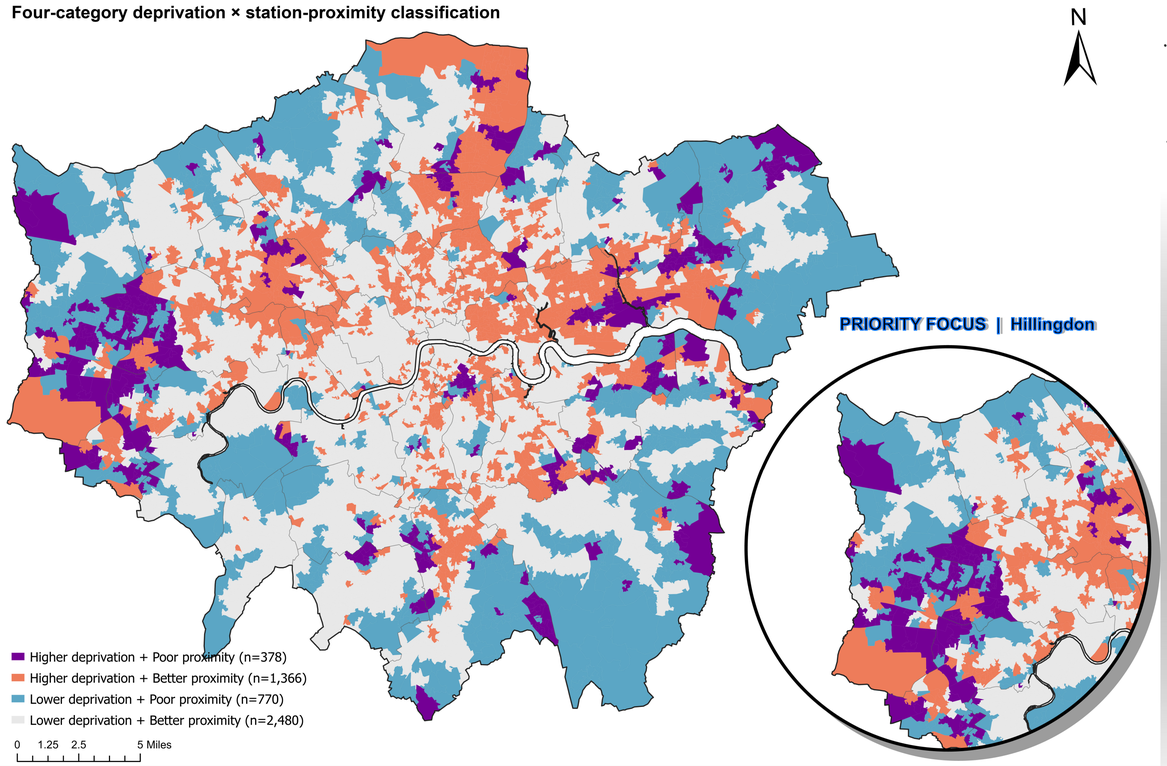
The deprivation and station-proximity classifications were combined into four bivariate groups. This identifies neighbourhoods where socioeconomic disadvantage and weaker rail proximity coincide, but it does not represent official PTAL or a complete measure of public-transport accessibility.

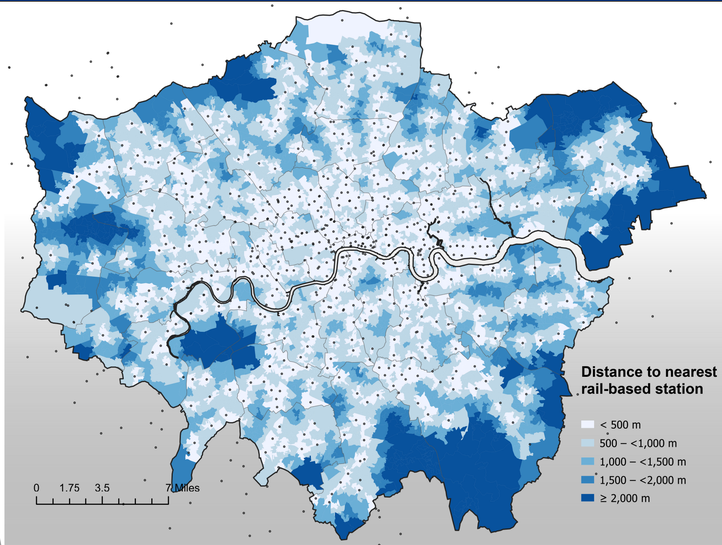
Methodological Workflow

Integration of IMD 2025 deprivation indicators with nearest rail-station proximity at LSOA level across London



Four-category deprivation × station-proximity classification





CARTOGRAPHIC PRINCIPLES & DESIGN

The poster uses a strong visual hierarchy, with the combined bivariate map presented as the dominant analytical output and the separate deprivation and station-proximity maps providing supporting spatial context. A restrained navy-and-neutral design frames the analysis, while purple is reserved to emphasise areas of higher socioeconomic deprivation and poor rail-station proximity. Borough boundaries remain visually subordinate to the LSOA-level patterns, while legends are positioned close to their associated maps. Consistent typography, spacing and map styling are used throughout the A1 layout to improve readability, support comparison between maps and maintain a coherent professional cartographic presentation.

LIMITATIONS & UNCERTAINTY

Nearest-station proximity is based on planar straight-line distance from one representative point per LSOA and does not represent actual walking routes.

The 1 km threshold is an analytical screening criterion, not an official accessibility standard; results are therefore sensitive to the selected threshold.

Station proximity does not account for service frequency, journey time, bus connectivity, affordability or individual mobility, while IMD represents area-level deprivation and does not imply causation.

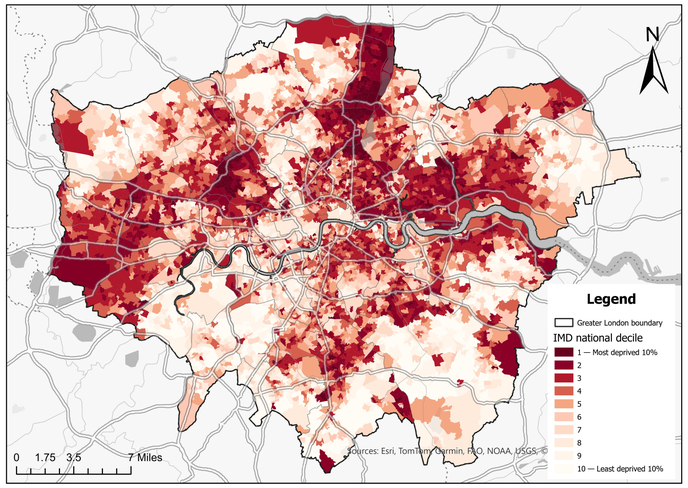
RESULTS & CRITICAL DISCUSSION

Of the 4,994 London LSOAs analysed, 1,744 (34.92%) fall within IMD national deciles 1–3, while 1,148 LSOAs (22.99%) are located at least 1 km from their nearest rail-based station. The key priority class contains 378 LSOAs (7.57%) where higher deprivation overlaps with poorer station proximity. The remaining categories comprise 1,366 higher-deprivation LSOAs with better proximity, 770 lower-deprivation LSOAs with poor proximity, and 2,480 lower-deprivation LSOAs with better proximity.

The spatial pattern indicates that weaker rail proximity is not uniformly associated with deprivation across London. Most higher-deprivation LSOAs remain within 1 km of a rail-based station, while a smaller but important group experiences both socioeconomic disadvantage and weaker rail proximity. Priority LSOAs occur in several parts of London, with notable concentrations in areas including Hillingdon, Ealing, and Barking and Dagenham. This suggests that the relationship between deprivation and rail access is geographically uneven rather than forming a simple inner–outer London pattern. Transport disadvantage may interact with wider socioeconomic disadvantage, making these overlapping areas particularly relevant for targeted accessibility assessment and place-based planning [4].

The results are also sensitive to the selected proximity threshold. Using a 750 m threshold increases the higher-deprivation/poor-proximity group to 636 LSOAs, while the adopted 1,000 m threshold identifies 378 LSOAs and a 1,250 m threshold reduces the group to 219 LSOAs. This confirms that the threshold materially affects the number of areas classified as poorly connected. The 1 km rule should therefore be interpreted as a transparent screening threshold rather than a definitive accessibility standard.

The findings should nevertheless be interpreted cautiously. Straight-line distance from one representative point per LSOA does not reflect actual walking routes, station entrances, service frequency, waiting time, journey time, interchange opportunities, affordability, bus connectivity, crowding or individual mobility constraints. TfL accessibility approaches incorporate additional service characteristics beyond distance alone [3]. IMD is also an area-based measure of relative deprivation and does not describe every individual resident. Consequently, the mapped overlap identifies spatial coincidence and potential priority areas, but it should not be interpreted as evidence that weaker rail access causes deprivation.



Conclusion / Key Message

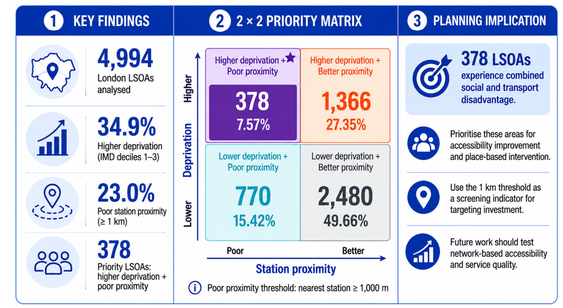
The analysis identified 378 LSOAs (7.57%) where higher deprivation overlaps with poor rail-station proximity of 1 km or more. These locations represent the clearest areas of combined socioeconomic and transport disadvantage within the study and therefore provide a useful basis for prioritising more detailed accessibility assessment.

The findings also show that deprivation and rail proximity do not follow a simple one-to-one relationship across London. Many higher-deprivation neighbourhoods remain relatively close to rail-based stations, while some lower-deprivation areas experience weaker station proximity. This reinforces the importance of analysing both dimensions together rather than relying on either deprivation or transport proximity alone.

The 1 km threshold should be treated as a screening criterion rather than a complete measure of accessibility. Future assessment should incorporate walking routes, station entrances, service frequency, journey time, interchange opportunities, bus connectivity and other dimensions of transport accessibility.

Planning message: The identified overlap areas should be prioritised for further investigation and place-based transport planning, particularly where improved connectivity could support access to employment, education and essential services.

WORD COUNT : ~1750 Words



REFERENCES

[1] Ministry of Housing, Communities and Local Government (MHCLG) (2025). English Indices of Deprivation 2025: Statistical Release. UK Government.

[2] Geurs, K.T. and van Wee, B. (2004). Accessibility evaluation of land-use and transport strategies: Review and research directions. Journal of Transport Geography, 12(2), 127–140.

[3] Transport for London (TfL). Assessing Transport Connectivity in London: Connectivity Assessment Guide. Transport for London.

[4] Lucas, K. (2012). Transport and social exclusion: Where are we now? Transport Policy, 20, 105–113.

[5] Office for National Statistics (ONS) (2021). Lower Layer Super Output Areas (December 2021) Boundaries EW BGC V5. ONS Open Geography Portal.

[6] Transport for London (TfL) (2026). Open Data / Unified API: Rail-based Station Locations. Transport for London. Data extracted 9 August 2026.

Data Sources: English Indices of Deprivation 2025 — Ministry of Housing, Communities & Local Government (MHCLG); 2021 Lower Layer Super Output Area boundaries — Office for National Statistics (ONS); rail-based station locations — Transport for London (TfL) Open Data / Unified API.
Coordinate Reference System: British National Grid (EPSG:27700) | TfL data extraction: 9 August 2026.

Methodological note: Nearest-station accessibility is represented by planar straight-line distance from an LSOA representative point and should not be interpreted as official PTAL or full public-transport accessibility.

AI-use acknowledgement: Generative AI was used only to support the development and design of supporting figures.

CRS: British National Grid (EPSG:27700) | TfL data extracted: 9 August 2026 | Distance method: Planar nearest-station distance |